

## Divide and Conquer recursion

1. Solve the following recurrences. State tight asymptotic bounds for each function in the form  $\Theta(f(n))$  for some recognizable function  $f(n)$ . Assume reasonable but nontrivial base cases if none are supplied.

(a)  $A(n) = 4A(n/3) + n^2$

(b)  $B(n) = 3B(3n/5) + n$

(c)  $C(n) = 4C(n-1) + 2$